

Random numbers

The “pick random” block can be used to select a random number between two values. This block is useful for rolling dice in a game or for when you want to mix up a sprite’s costumes.

pick random 1 to 10

You can change the numbers in the block

◁ Pick a random number

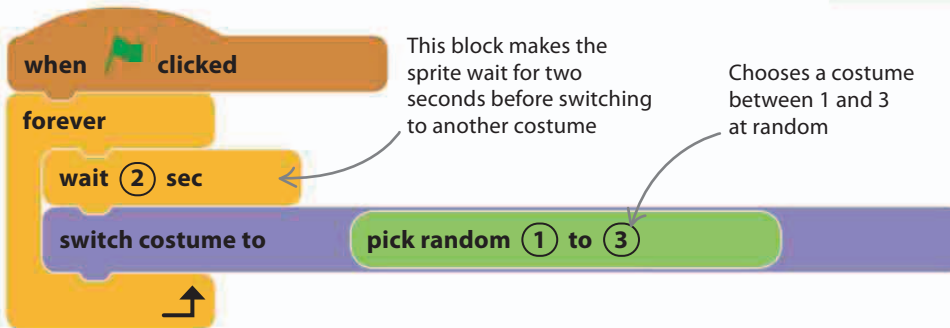
To pick a random month, change the numbers to choose a number between 1 and 12.



EXPERT TIPS

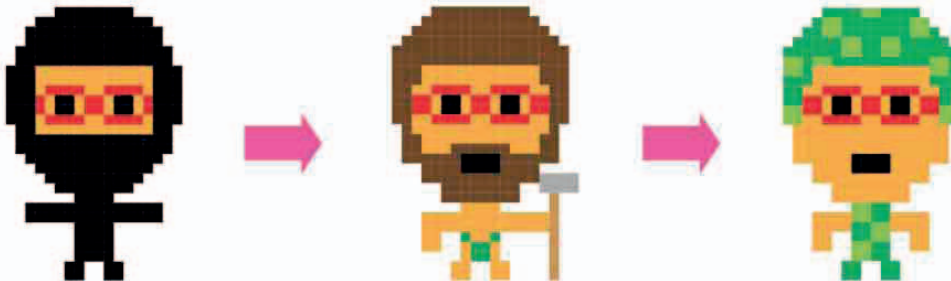
Gaming

Computers often use random numbers to add surprises to games. For example, an alien can be made to appear in a random place, or after a random amount of time. It can be used to select a random value, as if you were rolling a dice, or a random costume for a sprite.



◁ Switching costumes

This script changes a sprite’s costume at random every two seconds.



◁ Random costumes

Costumes can make a sprite appear to move its body, or might give it different clothes, as shown here.

Hard math

Simple “Operator” blocks can do most calculations, but Scratch can also do more complex math. The “mod” block divides two numbers and gives the remainder, which is the number that is left over. The “round” block rounds to the nearest whole number, and the “sqrt” block gives the square root of a number.

10 mod 3

Gives nearest whole number to 44.7

Divides 10 by 3 and gives the remainder—the number that is left over

round 44.7

Choose different functions from the drop-down menu

sqrt of 9

Calculates the square root of 9

◁ More math

The “Operator” section has blocks of advanced math functions that can be used to do complex calculations.